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Earth's Sky: Red Sky at Morning

Weather Lore and the Science Behind It

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Introduction

Every culture that works outdoors has a version of it, and the English-speaking world's version rhymes: red sky at night, sailors' delight; red sky at morning, sailors take warning. The saying compresses a piece of real atmospheric physics into fourteen syllables, and it does so accurately enough that meteorologists still teach the mechanism behind it. A sky burning red at sunrise is one of the most spectacular sights the atmosphere offers, and according to the proverb it is also one of the most useful, a visible sign that the weather is about to turn. This book is about that double life: the red morning sky as an optical event and as a forecasting tool, as a line of scripture and a line of poetry, as a phenomenon you can explain with an equation and one you can still stand outside and simply gawk at.

The physics turns out to be generous. Sunlight at daybreak enters the atmosphere sideways, grazing the curve of the Earth, and the long path strips the blue from it, leaving the red that survives to splash across whatever clouds and haze stand in its way. That much is optics, settled for a century and a half. The forecasting layer is dynamical: in the middle latitudes, weather systems tend to travel from west to east, so a sunrise that lights the sky from a clear eastern horizon while clouds bank in the west is, more often than not, announcing rain moving in. The proverb is a marriage of scattering and circulation, an observation of light and an inference about wind, and it is one of the very few pieces of folk meteorology that survives contact with the satellite era intact.

The phrase itself has an unexpectedly deep pedigree. The oldest widely cited version appears in the Gospel of Matthew, where Jesus, chiding listeners who read the sky but not the times, quotes the saying as an already proverbial piece of weather wisdom: a red evening sky promises fair weather, a red morning sky threatens foul. Two thousand years later the same observation appears in sailors' rhyme collections, farmers' almanacs, and the notebooks of atmospheric scientists. Between those two ends lie Shakespeare, the eruption of Krakatoa, the paintings of Turner, the invention of the theory of scattering, and a century of argument about whether the proverb is really as reliable as it feels.

This book follows the phenomenon the way the rest of this series follows its subjects. First the history: where the proverb came from, how it traveled, and who first tried to explain the colors it describes. Then the science: the scattering of light, the behavior of clouds at dawn, the strange optics of twilight. Then the practice: how observers and forecasters have used sky color, how well the proverb actually works, and where its predictive power runs out. And finally the extended family of the phenomenon: volcanic afterglows, the Belt of Venus, the green flash, blue Martian sunsets, and the other spectacles that share the red morning sky's physics.

A note on the word morning itself. The red sky of the title is specifically the sunrise sky, and sunrise differs from sunset in more than direction. The atmosphere at dawn is usually cleaner and more stable than at dusk, which makes the morning's reds sharper but rarer; the proverb treats morning red as the warning and evening red as the promise, and the atmosphere treats the two horizons differently too. This book keeps both in view, because the two halves of the day illuminate, literally, each other.

The red sky at morning, in short, is a phenomenon worth a whole book precisely because it sits on a seam: between folk knowledge and physical law, between art and optics, between the oldest recorded weather observation in common circulation and the newest satellite watching aerosols circle the planet. It is the oldest weather forecast most people know, and, used with an understanding of what it can and cannot say, still one of the best.

A word on what the red actually is, since it is easy to mistake the effect for the substance. The Sun emits the same colors at dawn as at noon; nothing about the light itself has changed overnight. What changes is the sampling. The atmosphere, which at noon takes a short vertical toll from the sunlight, at dawn takes a long grazing toll, and it collects that toll unequally, taking the blue far more readily than the red. A red morning sky is therefore not a different light but a different bill: the same spectrum, charged for a much longer passage, with the blue wavelengths footing most of the cost. Every chapter of this book is, one way or another, about how that bill is calculated, who pays it, and what the receipt can tell you about the weather to come.

Chapter 1: The Discovery of Red Sky at Morning

Humanity did not discover red skies; they are unmissable. What it discovered, over long centuries, was that the color meant something. The oldest widely cited statement of the idea is the one quoted in the Gospel of Matthew, in a passage where the speaker complains that his audience can read the sky but not the signs of their own

moment: when the sky is red and lowering in the morning, foul weather follows; when it is red in the evening, fair weather is on the way. The phrasing matters, because it presents the rule not as a new teaching but as an old one, something everyone in earshot already practiced. By the first century, in other words, the red-sky proverb was already a cliche, and the Gospel passage stands as the earliest surviving record of it rather than its origin.

The observation is old enough to be built into language. Greek epic poetry greets each dawn with an epithet that is essentially a color observation: Dawn is rosy-fingered, and the poet reaches for the same image every time day breaks, because the fingers of pink light spreading above the eastern horizon are the visible signature of sunrise in the clean Mediterranean air. That is a literary discovery of the same fact, made independently and made so early that it became a formula. When a culture's stock adjective for the goddess of morning is a description of atmospheric scattering, the culture has noticed its sky with some attention.

The proverb's seafaring career grew out of necessity. For a ship under sail, weather is not scenery but the difference between profit and loss, life and death, and sailors before instruments had every reason to squeeze forecasts out of every visible sign: the color of the water, the behavior of birds, the haloes around the moon, and, standing above most of these, the color of the sky at the day's two hinges. The red morning sky matters to a sailor because rain and rising wind, the weather it typically heralds in the middle latitudes, demand sail changes and watchfulness hours before they arrive; a forecast made at dawn buys a crew the morning to prepare. The proverb's English rhyming form, with sailors or shepherds taking delight or warning, is the form it took in the oral tradition of exactly those people whose livelihoods depended on the forecast being right.

The scientific discovery of the mechanism took much longer, because it required the discovery of what light is. Newton had shown that white light is a mixture of colors in the seventeenth century, but the sky's colors resisted explanation until the wave theory of light matured. The turning point came in 1869, when John Tyndall demonstrated in the laboratory that fine particles in air scatter blue light far more strongly than red, and in 1871, when Lord Rayleigh derived the law that bears his name: the scattering by molecules is inversely proportional to the fourth power of wavelength, so the shortest visible wavelengths are scattered an order of magnitude more strongly than the deepest reds. The morning sky was red for the same reason the daytime sky was blue, and both facts now had an equation. The proverb's optical half was solved; its dynamical half, the west-to-east travel of weather, followed as the science of the atmosphere's general circulation matured across the decades around the turn of the twentieth century.

The modern study of red skies received an unplanned natural experiment in 1883, when the volcano Krakatoa, in the Sunda Strait, exploded with a force heard thousands of kilometers away and injected sulfate aerosols into the stratosphere. For more than a year afterward, sunrises and sunsets around the world burned with colors so far beyond the ordinary that newspapers printed descriptions daily, observers in the tropics saw the Sun itself appear blue or green, and a reddish ring around the Sun was named after the Hawaiian pastor who reported it first. The Krakatoa afterglows taught atmospheric science something it had never had before: a global, datable event whose fingerprint in the sky could be studied systematically, and the published records of those sunsets became a foundation for the study of stratospheric aerosols, the same science that today measures volcanic influence on climate.

So the discovery of red sky at morning happened in stages, and each stage belongs to a different kind of discovery. The observation is prehistoric; the proverb is ancient; the mechanism is Victorian; the global calibration is a byproduct of the largest eruption in recorded history; and the modern assessment of the proverb's reliability belongs to the satellite age. Few subjects gather a first-century text, a line of Shakespeare, a laboratory experiment, and a stratospheric cloud of sulfuric acid droplets into one story.

The classical world systematized the reading of sky signs long before it could explain them. Theophrastus, Aristotle's successor at the Lyceum, compiled a treatise on weather signs that catalogued, among hundreds of other omens, the colors of the dawn as predictors, and his work joined a tradition of weather-lore poems, of which Aratus's Phaenomena, with its companion section on weather signs, was the most widely read: farmers across the Greek and Roman world learned the sky through verse, and the color of morning was a stock entry. Pliny the Elder preserved the same kind of rules in his encyclopedic Natural History, and the Latin and Greek tradition passed a working body of color forecasting down through the medieval almanac compilers into the age of print. The red morning sky has been an item in a professional literature, on and off, for over two thousand years.

The modern institutional history begins with the practical needs of the sea. The first national storm-warning service, Robert FitzRoy's Meteorological Office of the 1860s, grew directly out of the fishing disasters of the British coasts, and FitzRoy, a sailor himself who had commanded the Beagle of Darwin's voyage, filled his forecast manuals with

exactly the kind of sky-reading the red-sky proverb encodes, alongside the new instruments, barometers above all, that he lobbied to get into every fishing port. The proverb and the barometer worked the same beat, and the Victorian fisherman who consulted one was expected to know the other. The weather services that grew from that beginning gradually replaced the rhyme with the synoptic chart, but the chart, with its fronts and pressure systems moving west to east, ultimately vindicated the rhyme it replaced: the old observation had been tracking the same circulation all along.

Long before the mechanism was solved, the painters had documented the effect with an accuracy the physicists would later envy. Leonardo da Vinci, whose notebooks treat the sky as an optical problem, wrestled with the haze at the horizon, correctly attributing the whitening of distant air to intervening atmosphere rather than to the objects themselves, an idea centuries ahead of its formalization; his distant mountains painted blue with distance are, in effect, scattering measurements in egg tempera. The Dutch landscape school recorded the low Sun's color and the structure of evening and morning skies with the eye of witnesses, and the later Turner, painting through the volcanic 1810s, delivered sunrises so chromatically extreme that researchers eventually used them as proxy data for stratospheric aerosol. Art was the subject's observational archive for the centuries before instruments, and it is still, as the earlier chapters showed, occasionally consulted as one.